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Integrated planning for electric commercial vehicle fleets: A case study for retail mid-haul logistics networks

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ABSTRACT

Electric commercial vehicles can contribute significantly to sustainable transportation. However, they are still perceived as being less economically viable than internal combustion engine vehicles. To this end, we analyze the deployment of electric commercial vehicles in retail mid-haul logistics fleets for current and future technology scenarios. We develop a novel assessment methodology that combines total cost of ownership calculations with a rich location-routing model. We consider integrated strategic network design and operational routing decisions over a multi-period time horizon, accounting for mixed fleets of electric and conventional vehicles and battery degradation. To solve this problem, we develop a novel matheuristic that embeds a state-of-the-art metaheuristic to take operational routing decisions. With this framework, we analyze a possible transition towards electrified logistics fleets. Starting with a real-world case, we perform sensitivity analyses based on a technology roadmap and further network structures. We discuss cost structures, the time-dependent nature of investment decisions, operational characteristics, and potential emissions savings. We show that in certain cases an electrification of mid-haul logistics fleets is operationally feasible, economically viable, and provides environmental advantages.

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1. Introduction

Electric vehicles constitute a promising solution for the reduction of noise and polluting emissions. Currently, nearly half of the transport-related CO₂ emissions results from freight transportation, and forecasts point to a 160% increase of CO₂ emissions related to transport as total freight demand is projected to triple by 2050 (OECD, 2017). Moreover, impending bans of internal combustion engine vehicles (ICEVs) from city centers call for the deployment of electric commercial vehicles (ECVs) in logistics fleets as a prerequisite to the attainment of reduction thresholds on NO₂ and soot (IEA, 2018).

While the number of ECVs in logistics fleets is still very low, recent positive experiences in pilot projects have accelerated the diffusion of light-duty ECVs in city and short-haul logistics. For instance, the Deutsche Post DHL group (DPDHL) tested a first elec-

tric fleet of 12 Renault Kangoo ZE for postal services in German cities as early as 2011 (DPDHL, 2011). A larger pilot project was launched in 2013, aiming at zero-emission deliveries for an urban region with 310,000 inhabitants. Within this project, custom-built ECVs were used (DPDHL, 2013; 2014b; Pieringer, 2013), and DPDHL eventually bought the company that manufactured the ECVs as a reaction to the project's success (DPDHL, 2014a). Other major players such as FedEx and UPS followed this trend (FedEx, 2018; UPS, 2018). The driving range of today's light-duty ECVs now suffices for most trips in city logistics, whose average length is below 80km. This means that ECVs are becoming a viable option in this context. A sufficient driving range remedies the ECVs operational constraint and allows logistics service providers (LSPs) to consider a transition from a conventional to an electrified fleet, based on the total cost of ownership (TCO), which corresponds to their business practice.

Yet LSPs are reluctant to electrify their fleets in mid-haul logistics because several obstacles remain. Indeed, in most mid-haul applications, the driving range of medium-duty ECVs is not sufficient to avoid time-consuming recharging processes during daily operations. Often, these recharges are even necessary between

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customer visits. Over time, this effect worsens further, as batteries degrade, which results in an even lower battery capacity and hence in decreasing driving ranges, yielding increasing operational difficulties. In parallel to solving these operational problems, the LSP has to decide on additional investments in charging infrastructure to ensure reliable recharging within the network. Besides increasing investments, these decisions also bear an inherently complex interdependency because fleet operations and charging infrastructure investments are complementary (Schiffer & Walther, 2017). Furthermore, the impact of these obstacles may vary significantly with respect to a mid-haul network's characteristics or technological specifications, and to developments in ECVs and charging infrastructure.

In order to foster the use of ECVs in mid-haul logistics, the German government has initiated and partly funded a pilot project to evaluate the competitiveness of ECVs compared with that of ICEVs, and to determine whether ECVs can be integrated in existing mid-haul logistics fleets (cf. Stütz et al., 2016). Herein, several logistics fleet operators (for instance the retail company TEDI that constitutes our case study) used ECVs for their delivery processes. The results of this pilot project indicate that ECVs can be operated and may be on the verge of breaking even in the context of mid-haul logistics. However, a chicken and egg dilemma remains. In field work, several LSPs stated that they are reluctant to make strategic decisions on a TCO basis because the level of information provided does not allow them to understand all operational implications. Instead, they ask for a proof of concept that ensures the operational feasibility of their fleet under the investment decisions taken. Providing such a proof of concept via scientific analyses remains a challenging and contestable task since mid-haul logistics networks are typically affected by uncertain customer patterns (cf. Schiffer & Walther, 2018b), and one may question the accuracy of integrated planning approaches that combine these uncertainties with integrated strategic network design decisions, operational analysis, and TCO analysis.

Mid-haul logistics networks operated by retail companies in order to supply their own stores, remain an exception. In these networks, the delivery volume and the opening of new stores (customers) is centrally planned by the retail company and remains stable over a short to medium planning horizon. For example, in our case study (see Section 4) the demand of each store remains stable but the demand in between stores varies with a standard deviation of $\sigma = 2.06$ in an interval of $[4, 12]$ pallets per week. Hence, we see retail mid-haul logistics networks as a natural starting point to develop integrated evaluation frameworks for electrified mid-haul logistics. Integrated analyses are necessary but uncertain customer patterns are subordinate in this context, which helps to reduce the complexity of the analysis.

Against this background, we develop the first integrated evaluation framework that combines TCO analysis with strategic network design decisions, as well as an operational proof of concept. We explicitly consider both the long-term transition process towards an electrified fleet, i.e., the replacement of vehicles and the installation of charging infrastructure, as well as the short-term operation of the fleet, i.e., the fulfillment of customer demands under operational limitations with respect to range, recharging times, and battery degradation. This framework allows to carry out the first integrated study on viable deployment scenarios for ECVs in retail mid-haul logistics, considering developments in costs and technologies over time. Our results yield tangible transformation strategies with which we aim to foster the adoption of medium-duty ECVs in retail mid-haul transportation and to unleash the emission savings potential in this sector.

To set our study apart from recent research, we first analyze the related literature in Section 1.1, before detailing the scope and structure of this paper in Section 1.2.

1.1. Related literature

Recent research on ECVs can be divided into two streams focusing on (i) aggregated cost analyses with TCO calculations to investigate the competitiveness of ECVs and (ii) models providing decision support on both integrated network design and fleet operations, based on location-routing problems (LRPs) and vehicle routing problems (VRPs). In the following, we review both streams concisely.

Aggregated cost analyses

To evaluate the competitiveness of ECVs, it is common to perform aggregated TCO calculations in order to compare the life-cycle costs of ECVs and ICEVs. Usually, these analyses rely on general assumptions regarding operational parameters, such as average total distances instead of specific route lengths. Some analyses try to overcome these shortcomings, but still rely on static approximations. Lee, Thomas, and Brown (2013) conducted a TCO calculation for medium-duty ECVs under realistic energy consumption values and driving cycles within a range from 48 to 96 kilometers. Davis and Figliozzi (2013) conducted a TCO analysis based on fuel consumption, as well as approximate routing constraints, battery replacement, and speed profiles. Taefi, Stütz, and Fink (2017) provided a TCO analysis of ECVs focusing on the cost-optimal balance between a high vehicle utilization and the resulting increase in required battery replacement caused by battery degradation. All of these analyses find that ECVs become more competitive with increased traveled distance. Furthermore, characteristics of specific driving patterns such as frequent stops, congestion, engine idling, and low speed increase the competitiveness of ECVs.

While these analyses rely on reasonable assumptions, they are not sufficient to alleviate the skepticism of practitioners, since they consider neither operational limitations nor case-specific restrictions. For example, in the Deutsche Post DHL group, ECVs became competitive for postal services for significantly smaller daily mileages than was anticipated by general TCO assessments. Hence, doubts on the results of aggregated TCO assessments, i.e., over- or underestimations of costs, seem to be justified. Moreover, all existing studies, analyze a static scenario with investment decisions taken at one single point in time, i.e., at the beginning of the planning horizon. Such an experimental design fails to analyze the dynamic transformation of fleets and does not account for changes in prices and technologies over time. Focusing specifically on mid-haul operations, existing modeling approaches do also not allow to capture costs for charging infrastructure appropriately, as analyzing the need for charging infrastructure requires a detailed modeling of daily fleet operations.

Vehicle routing and location-routing problems

In the transportation sector, analytical tools are commonly used to make daily planning tasks more efficient and profitable. In this context, various kinds of VRPs have been proposed to address the operational planning of pickup and delivery networks for conventional logistics fleets. These models have recently been extended by incorporating constraints for ECV logistics fleets, i.e., recharging operations and limited driving ranges. In the following, we focus on major contributions and refer for extensive overviews to Pelletier, Jabali, and Laporte (2016) and to Schiffer, Schneider, Walther, and Laporte (2019) for goods distribution, and to Pelletier, Jabali, Laporte, and Veneroni (2017) for battery behavior.

Conrad and Figliozzi (2011) presented the recharging VRP, considering battery capacity limitations of ECVs and charging opportunities at customer vertices. Erdoğan and Miller-Hooks (2012) introduced the first model that considers additional vertices allowing for charging activities for any kind of alternative fuel vehicle (AFV). Schneider, Stenger, and Goeke (2014) developed the electric VRP (EVRP) with time windows (EVRP-TW), the first model that explicitly focuses on ECVs. Later publications then

extended the basic EVRP-TW by considering heterogeneous fleets (Goeke & Schneider, 2015; Hiermann, Puchinger, Ropke, & Hartl, 2016), partial recharging (Desaulniers, Errico, Irnich, & Schneider, 2016; Felipe, Ortuno, Righini, & Tirado, 2014; Montoya, Gu  ret, Mendoza, & Villegas, 2017), charging stations with different charging rates and costs (Felipe et al., 2014; Montoya et al., 2017), non-linear charging functions (Froger, Mendoza, Jabali, & Laporte, 2019; Montoya et al., 2017), hours of service regulations (Schiffer, G., Schneider, & Walther, 2017), and shared charging stations (Ko  , Jabali, Mendoza, & Laporte, 2018). The first exact solution methods were presented by Desaulniers et al. (2016); Hiermann et al. (2016), and Roberti and Wen (2016). In addition, the first papers that simultaneously consider charging station location and vehicle routing decisions were published as variants of the LRP with intra-route facilities (LRPIFs) (Schiffer, Schneider, & Laporte, 2018; Schiffer & Walther, 2017; 2018a; 2018b; Yang & Sun, 2015). VRP models that account for specific characteristics of retail logistics, like driver shifts, delivery over several time periods (e.g., working days of one week), as well as combined pickup and delivery processes, were considered for ICEVs. In this context, methodologies for ECVs and LRPIFs are still needed.

The methodological contribution of these papers is to extend existing problem classes or to develop new algorithms that are typically tested on artificial instances. No comparative competitiveness analysis between ECVs and ICEVs has yet been performed based on these models. A recent study by Arroyo, Felipe, Ortuno, and Tirado (2019) combined a TCO analysis with a green VRP in order to study the impact of carbon pricing on the competitiveness of ECVs. However, this study focuses on a static investment decision at one point in time and neglects charging station investments. Additionally, none of the existing models accounts for features such as driver shifts or multi-day deliveries in the case of ECVs, although such models are available for ICEVs. Finally, with the possible exception of Pelletier, Jabali, and Laporte (2018) who address battery degradation within a setting of fixed routes, publications on battery degradation effects have so far neglected the routing decisions.

Concluding, both aggregated TCO analyses and operations research models for the planning of electric mid-haul logistics fleets have not yet been able to establish a fair comparison between ECVs and ICEVs. Besides lacking specific problem characteristics, the existing approaches neither analyze the transformation of a logistics fleet with respect to its investment decisions over a dynamic time horizon regarding changing cost and technology parameters, nor do they model the operations of a partly electrified logistics fleet under real-world requirements.

1.2. Scientific contributions and organization

To overcome these limitations, we develop a new integrated methodological framework that allows for the assessment and identification of deployment strategies for ECVs in the sense of a transformation of conventional logistics fleets towards electrified fleets. We consider strategic investments and operational costs from a fleet operator's TCO perspective. Our model accounts for long-term strategic investments with regard to network and fleet transformation over multiple periods, as well as for anticipated short-term fleet operational costs based on routing decisions, while accounting for potential ECV limitations.

This paper makes several scientific contributions. First, we develop a new integrated modeling concept merging a comprehensive TCO analysis with an integrated model that combines endogenous investment decisions on network and fleet design with fleet operations over a multi-period time horizon. Hence, we can assess the transformation of a logistics fleet into an electrified fleet, and we can adequately evaluate the competitiveness of ECVs

against ICEVs. Second, we enhance this model to consider ECV specific characteristics that affect fleet operations, such as limited driving ranges, battery degradation, and en route recharging. Third, we develop a novel metaheuristic based on a guiding scheme which overcomes the challenge of solving such a comprehensive problem. This scheme allows us to devise a branch-and-prune algorithm to make all investment decisions optimally while embedding a state-of-the-art metaheuristic component as an oracle to approximate the fleets daily operations. Fourth, we identify promising deployment scenarios for ECVs in mid-haul logistics to foster the transformation of logistics fleets in practice. To this end, we apply our algorithmic framework to the case of a large German retail company. Fifth, we enhance our findings beyond this specific case by carrying out sensitivity analyses for technological and cost parameters, but also for the characteristics of the logistics network. We discuss all results with respect to overall costs for ECV and ICEV fleets. Additionally, we analyze the potential emissions savings arising from such a fleet transformation in order to quantify the ecological benefit of deploying ECVs. Finally, we derive managerial insights on the current and future viability of ECVs in retail mid-haul logistics fleets, and we provide findings of interest to practitioners and researchers.

The remainder of this paper is structured as follows. We first introduce our problem setting in Section 2 and develop our methodology in Section 3. We then describe our case study and the overall experimental design in Section 4. Section 5 presents results on the transformation of logistics fleets towards ECVs from an economic and environmental perspective for our real-world case, and provides the results of our sensitivity analyses. Section 6 concludes this paper by summarizing its main findings and deriving managerial insights.

2. Problem definition

This section introduces our planning problem. First, we motivate the planning problem that LSPs face in retail mid-haul logistics in Section 2.1. Then, we formalize our planning problem which integrates a TCO analysis with all associated strategic and operational requirements and interdependencies as a mixed integer program (MIP) in Section 2.2.

2.1. Problem setting

In retail mid-haul logistics, a (retailer-owned) LSP operates a fleet of medium sized (e.g., 12.5-tonne) trucks to deliver goods to customers while meeting additional operational constraints, e.g., customer delivery time windows. Whereas such a general setting is common to the well-known vehicle routing problem, these retail mid-haul logistics networks bear some specific characteristics:

1. Retail mid-haul logistics networks are far less dynamic than short-haul or city logistics networks for last-mile services, because the retail company plans the delivery volume of each store in advance and aims at constant volumes. Indeed, the operations are often executed in a repetitive fashion, as it is also the case in our real-world example (cf. Section 4.1). Accordingly, the demand structure of mid-haul logistics networks can be much better estimated, even over a long horizon.
2. Although demand patterns are relatively constant, other characteristics of retail mid-haul logistics network properties such as customer time windows, the spatial distribution of customers, and the granularity of the customer demand may vary significantly. These properties influence the average number of customers a vehicle can serve per route. On an operational level, the number of operated working shifts per day may vary as well.

Some of these characteristics work in favor of our analyses, while others complicate our experimental design. On the one hand, stable or well estimated demand patterns over time ease the mid-term analysis of such a network as they allow for a medium granular resolution in the time dimension. On the other hand, the multitude of varying network properties does not allow to draw general conclusions based on a single representative case study, but requires extensive sensitivity analyses to cover as many application cases as possible.

Additional constraints arise if the LSP decides to operate ECVs. First, ECVs have a limited driving range and require time-intensive recharging to replenish the battery's state of charge. Moreover, the capacity of the battery decreases over time, resulting in a further diminished driving range. Since the size of a mid-haul logistics network may exceed the driving range of current ECVs, en route recharging may be necessary. Besides considering recharging stops at the operational level, this requires the installation of a charging station network at the strategic level. While one may assume that an LSP could rely on public charging infrastructure, most LSPs will prefer to establish their own charging infrastructure on private or semi-private grounds protected from vandalism and blocking cars to ensure reliable operations.

In this setting, the LSP faces planning tasks at the operational and at the strategic level:

1. At the operational level, the LSP decides on daily route plans to deliver goods to customers. As is common in retail mid-haul logistics, these route plans are repeatedly used on a weekly or monthly schedule. If the LSP operates ECVs, it must additionally schedule recharging operations. In practice, metaheuristics are used to conduct these planning tasks.
2. At the strategic level, the LSP decides on fleet investments, i.e., when to replace a vehicle. For ICEV fleets that have no operational limitations, LSPs make such decisions based on TCO calculations. If ECVs are used, the LSP must additionally determine charging infrastructure investments, considering interdependencies with daily operations and vehicle deployment decisions. Neglecting these interdependencies endangers the feasibility of daily operations. Accordingly, mere TCO calculations are no longer sufficient to capture all relevant decision aspects in such a setting.

In the following, we provide a framework that allows for such a proof of concept by combining and enhancing existing operational planning components and strategic TCO analyses.

2.2. Mixed integer problem formulation

Our planning approach focuses on a logistics fleet operator's investment decisions to transform a logistics fleet towards ECVs within a given planning horizon $\mathcal{T} = \{1, \dots, n\}$. We assume that the fleet operator makes an investment decision for each vehicle to set up a mixed fleet of ICEVs and ECVs at time step $t = 0$. At every consecutive time step, she may make further investment decisions to replace ICEVs by ECVs, and to install ECV charging infrastructure. To prevent the bias arising from the fixed length of the planning horizon, we account for salvage values.

To assess these investment decisions accurately, we need to consider strategic decisions on network and fleet design but also their implications for operational routing decisions such that our planning problem belongs to the class of LRPIFs. However, existing modeling approaches are so far limited to segregated fleets of either ICEVs or ECVs and a static point in time. To overcome these limitations and to provide an approach that is suitable to derive decision support for practitioners, we develop the first LRPIF that accounts for multiple planning periods, mixed fleets, battery degradation, ECV limitations, and specific characteristics of the

logistics network like customer time windows and multiple driver shifts. In particular, the extension to multiple planning periods is highly important to assess real-world applications as accurately as possible. Although mid-haul logistics networks exhibit relatively stable spatial and temporal customer patterns, demands may vary over a larger time horizon. Our multiple planning period time horizon allows for such demand variations and even for spatial shifts, by modeling zero demands. Besides, we are the first to show how such a modeling approach can be combined with a TCO objective function to represent the economic impact of the fleet transformation.

Before we introduce our notation and the overall model, a specification on the time horizon $\mathcal{T}$ is in order. When combining strategic and operational decisions, the granularity of the time steps between these different decisions may vary significantly. In our specific case of retail mid-haul logistics networks, delivery schedules are executed on a repetitive schedule. Accordingly, we treat each time step $t \in \mathcal{T}$ as a time step in which the LSP can take an investment decision. To account for multiple delivery shifts or a pattern that extends over multiple days, we use depot copies with disjoint time windows, e.g., for a delivery pattern of five days with two daily shifts, we use 10 artificial time window disjoint depot nodes, each time window chosen such that it matches the customer time windows that belong to a specific shift. In the same vein, we solve the different resolution of cost factors. We keep the investment cost factors at the granularity of the time horizon $\mathcal{T}$ and scale operational cost factors accordingly, e.g., if a repetitive schedule occurs 12 times in between two investment time steps, we scale the operational cost factors by 12 accordingly.

In the remainder of this section, we index time-dependent parameters and variables with index t without further notion to keep the paper concise.

Notation: Our notation, as summarized in Table 1 holds as follows: we define our MIP on a directed graph $G = (\mathcal{V}, \mathcal{A})$ with a set of arcs $\mathcal{A}$ and a set of vertices $\mathcal{V} = \mathcal{C} \cup \mathcal{F} \cup \mathcal{F}' \cup \mathcal{D}^- \cup \mathcal{D}^+$. The set of vertices consists of several subsets: each vertex $v \in \mathcal{V}$ can be a customer vertex ($v \in \mathcal{C}$) or a vertex at which potentially a charging station vertex is allowed to be built ($v \in \mathcal{F}$). Note that $\mathcal{C}$ and $\mathcal{F}$ are not necessarily disjoint as in some cases, charging stations can be built at customer locations. To allow for multiple visits to charging stations, a set of dummy vertices $\mathcal{F}'$ is used and partitioned into subsets $\mathcal{F}'_\kappa$, each containing dummy vertices for each real vertex κ . Further, we use a set $\mathcal{D}^-$ and a set $\mathcal{D}^+$ of depot start and end vertices, which represent a single physical depot but are used to model the beginning and the end of routes. If we model a single work shift, $\mathcal{D}^-$ and $\mathcal{D}^+$ are singletons; if we model multiple work shifts, $\mathcal{D}^-$ and $\mathcal{D}^+$ contain multiple vertices (one for each shift) with disjoint time windows. Here, we remove arcs from $\mathcal{A}$ that may connect nodes between different shifts. For any $B \subseteq \mathcal{V}$, we define the sets $\delta(B) = \{(i, j) \in \mathcal{A} : i \in B, j \in B\}$, $\delta^-(B) = \{(i, j) \in \mathcal{A} : i \in B, j \notin B\}$, and $\delta^+(B) = \{(j, i) \in \mathcal{A} : j \notin B, i \in B\}$. Recall, that we use $\mathcal{T} = \{1, \dots, n\}$ to denote a finite time horizon with n discrete time steps. Since the complete model is time dependent, we index all time-dependent variables and parameters with t .

Let $\mathcal{K}$ be the set of vehicles for a given fleet size $|\mathcal{K}|$. We consider a homogeneous freight capacity F for all vehicles. We keep track of a vehicles remaining delivery freight upon arrival at vertex i by f_{it} and denote a customer's demand by p_{it} .

If the vehicle is an ECV, Q denotes its maximum battery capacity, i.e., its battery capacity in the first usage period, which has not yet been affected by battery degradation. With each time period, the battery capacity degrades by a percentage d of its initial capacity. We use q_{kt} to denote vehicle k 's battery capacity in time period t . Further, v_{it} denotes a vehicle's battery state of charge at vertex i , w_{it} denotes the amount of energy recharged at vertex i , and r denotes the amount of time necessary to recharge one unit

Table 1
Notation used in the MIP formulation.

$\mathcal{A}$	set of arcs (i, j)	q_{kt}	battery capacity of vehicle k
$\mathcal{V}$	set of vertices $(\mathcal{V} = \mathcal{C} \cup \mathcal{F} \cup \mathcal{F}' \cup \mathcal{D}^- \cup \mathcal{D}^+)$	v_{it}	vehicle's state of charge at vertex i
$\mathcal{C}$	set of customer vertices	w_{it}	amount of energy recharged at vertex i
$\mathcal{F}$	set of potential charging station vertices	h_{ijt}	energy consumption on arc (i, j)
$\mathcal{F}'$	set of dummy vertices	f_{it}	remaining freight upon arrival at vertex i
$\mathcal{F}'_{\kappa}$	set of dummy vertices linked to $\kappa \in \mathcal{F}$	p_{it}	delivery demand of customer i
$\mathcal{D}^-$	set of depot start vertices	τ_{it}	start time of service/recharging at vertex i
$\mathcal{D}^+$	set of depot end vertices	s_{it}	service time of customer i
$\delta(\mathcal{B})$	cut set denoting all arcs in $\mathcal{B}$	e_{it}	earliest start time of service at vertex i
$\delta^-(\mathcal{B})$	cut set denoting all arcs leaving $\mathcal{B}$	l_{it}	latest start time of service at vertex i
$\delta^+(\mathcal{B})$	cut set denoting all arcs entering $\mathcal{B}$	t_{ijt}	driving time on arc (i, j)
$\mathcal{T}$	set of time steps	z_{kt}^{ecv}	binary: vehicle k is replaced by an ECV
$\mathcal{K}$	set of vehicles	z_{k1}^{icev}	binary: vehicle k is replaced by an ICEV
F	vehicle freight capacity	y_{it}	binary: a charging station is built at i
Q	initial battery capacity of vehicle k	x_{ijkt}^{ecv}	binary: arc (i, j) is traversed by vehicle k
d	battery degradation rate	x_{ijkt}^{icev}	binary: arc (i, j) is traversed by vehicle k
r	refueling rate		

Table 2
Cost coefficients used in the objective function.

c_t^{ecv}	investment cost for one ECV	$c_{ijt}^{\text{ecv},o}$	operational cost for ECVs
c_1^{icev}	investment cost for one ICEV in $t = 0$	$c_{ijt}^{\text{icev},o}$	operational cost for ICEVs
c_{it}^{stat}	investment cost for one charging station	c_t^{driver}	driver costs

of energy. For each arc (i, j) , h_{ijt} denotes its energy consumption if it is traversed.

To model time dependencies, τ_{it} denotes the time at which a vehicle starts service or recharging at vertex i . Furthermore, s_{it} denotes the time it takes to provide service, i.e., deliver goods at customer i , and $[e_{it}, l_{it}]$ denotes for each customer $i \in \mathcal{C}$ the time window in which service has to start. If i is a depot vertex or a charging station vertex, $[e_{it}, l_{it}]$ denotes the time window in which a vehicle is allowed to arrive. We note that time windows at charging stations are not mandatory. In our specific case, we use depot time windows for separation purposes in case of modeling multiple delivery shifts. For other application cases, one may apply time windows to charging stations in order to model limited (time-dependent) access. For each arc (i, j) , t_{ijt} denotes the travel time a vehicle needs if it traverses (i, j) .

We use binary variables z_{kt}^{ecv} to indicate whether vehicle k is replaced by an ECV in period t ($z_{kt}^{\text{ecv}} = 1$) or not ($z_{kt}^{\text{ecv}} = 0$). Analogously, z_{k1}^{icev} denotes whether vehicle k is replaced by an ICEV in period 1 ($z_{k1}^{\text{icev}} = 1$) or not ($z_{k1}^{\text{icev}} = 0$). Note that, c_1^{icev} and z_{k1}^{icev} are only defined for $t = 1$, as we focus on a diffusion of ECVs in succeeding time periods. We indicate with y_{it} whether a charging station is built in or has been built prior to period t ($y_{it} = 1$) or not ($y_{it} = 0$). Binaries x_{ijkt}^{ecv} and x_{ijkt}^{icev} indicate whether arc (i, j) is traversed in period t by vehicle k as an ECV ($x_{ijkt}^{\text{ecv}} = 1$) or ICEV ($x_{ijkt}^{\text{icev}} = 1$) or not ($x_{ijkt}^{\text{icev}} = x_{ijkt}^{\text{ecv}} = 0$).

Objective: A TCO analysis for the operation of logistics networks considers different types of costs over time (2.1), i.e., one-time investments Inv_t , salvage values Sal_t , fixed costs Fix_t , and operational costs $Oper_t$ at each time step t , discounted at a rate ρ :

$$TCO = Inv + Sal + Fix + Oper = \sum_{t \in \mathcal{T}} \frac{Inv_t}{(1 + \rho)^t} + \frac{Sal_t}{(1 + \rho)^t} + \frac{Fix_t}{(1 + \rho)^t} + \frac{Oper_t}{(1 + \rho)^t}. \quad (2.1)$$

We transform (2.1) into a specific objective function for our planning task (2.2) using specific cost coefficients (see Table 2). We consider discounted investment costs for ECVs (c_t^{ecv}), for ICEVs (c_1^{icev}), and for charging stations (c_{it}^{stat}). Note that for the sake of conciseness, we include fixed costs (e.g., taxes) and salvage values

directly in the investment costs. Operational costs result for ECVs ($c_{ijt}^{\text{ecv},o}$), ICEVs ($c_{ijt}^{\text{icev},o}$), and drivers (c_t^{driver}). While operational costs are distance-based, we assume that a driver is always paid for a full shift and thus, we account for the respective cost for each depot-leaving route. With these notation, our objective results by considering costs at each time step $t \in \mathcal{T}$:

$$\begin{aligned} \min Z = & \sum_{k \in \mathcal{K}} c_1^{\text{icev}} z_{k1}^{\text{icev}} \\ & + \sum_{t \in \mathcal{T}} \left(\sum_{k \in \mathcal{K}} c_t^{\text{ecv}} z_{kt}^{\text{ecv}} + \sum_{i \in \mathcal{F}, t > 1} c_{it}^{\text{stat}} (y_{it} - y_{it-1}) + \sum_{i \in \mathcal{F}} c_1^{\text{stat}} y_{i1} \right. \\ & + \sum_{k \in \mathcal{K}, (i,j) \in \mathcal{A}} c_{ijt}^{\text{ecv},o} x_{ijkt}^{\text{ecv}} + \sum_{k \in \mathcal{K}, (i,j) \in \mathcal{A}} c_{ijt}^{\text{icev},o} x_{ijkt}^{\text{icev}} \\ & \left. + \sum_{(i,j) \in \delta^-(\{\mathcal{D}^-\})} c_t^{\text{driver}} (x_{ijkt}^{\text{ecv}} + x_{ijkt}^{\text{icev}}) \right). \end{aligned} \quad (2.2)$$

Here, we note that c_1^{icev} and z_{k1}^{icev} are only defined for $t = 1$, as we focus on a diffusion of ECVs in succeeding time periods.

Constraints: To cover all operational and strategic requirements, we state the following constraints:

$$\sum_{(i,j) \in \delta^-(\{i\}), k \in \mathcal{K}} x_{ijkt}^{\text{ecv}} + x_{ijkt}^{\text{icev}} = 1 \quad i \in \mathcal{C}, \quad t \in \mathcal{T} \quad (2.3)$$

$$\sum_{(i,j) \in \delta^-(\{i\}), k \in \mathcal{K}} x_{ijkt}^{\text{ecv}} + x_{ijkt}^{\text{icev}} \leq 1 \quad i \in \mathcal{V} \setminus \{\mathcal{C} \cup \mathcal{D}^- \cup \mathcal{D}^+\}, \quad t \in \mathcal{T} \quad (2.4)$$

$$\begin{aligned} & \sum_{(j,i) \in \delta^+(\{i\})} x_{ijkt}^{\text{ecv}} - \sum_{(i,j) \in \delta^-(\{i\})} x_{ijkt}^{\text{ecv}} \\ & = 0 \quad i \in \mathcal{V} \setminus \{\mathcal{D}^- \cup \mathcal{D}^+\}, \quad k \in \mathcal{K}, \quad t \in \mathcal{T} \end{aligned} \quad (2.5)$$

$$\begin{aligned} & \sum_{(j,i) \in \delta^+(\{i\})} x_{ijkt}^{\text{icev}} - \sum_{(i,j) \in \delta^-(\{i\})} x_{ijkt}^{\text{icev}} \\ & = 0 \quad i \in \mathcal{V} \setminus \{\mathcal{D}^- \cup \mathcal{D}^+\}, \quad k \in \mathcal{K}, \quad t \in \mathcal{T} \end{aligned} \quad (2.6)$$

$$0 \leq f_{jt} \leq f_{it} - p_{it}(x_{ijkt}^{\text{ecv}} + x_{ijkt}^{\text{icev}}) + F(1 - x_{ijkt}^{\text{ecv}} - x_{ijkt}^{\text{icev}})(i, j) \in \mathcal{A}, k \in \mathcal{K}, t \in \mathcal{T} \quad (2.7)$$

$$f_{0t} \leq Ft \in \mathcal{T} \quad (2.8)$$

$$\tau_{jt} \geq \tau_{it} + (t_{ijt} + s_{it})(x_{ijkt}^{\text{ecv}} + x_{ijkt}^{\text{icev}}) - l_{0t}(1 - x_{ijkt}^{\text{ecv}} - x_{ijkt}^{\text{icev}})(i, j) \in \delta^-(\mathcal{C} \cup \mathcal{D}^-), k \in \mathcal{K}, t \in \mathcal{T} \quad (2.9)$$

$$\tau_{jt} \geq \tau_{it} + t_{ijt}x_{ijkt}^{\text{ecv}} + r w_{it} - (l_{0t} + rQ)(1 - x_{ijkt}^{\text{ecv}})(i, j) \in \delta^-(\mathcal{V} \setminus \{\mathcal{D}^- \cup \mathcal{D}^+\}), k \in \mathcal{K}, t \in \mathcal{T} \quad (2.10)$$

$$e_{it} \leq \tau_{it} \leq l_{it}i \in \mathcal{V}, t \in \mathcal{T} \quad (2.11)$$

$$0 \leq v_{jt} \leq v_{it} + w_{it} - h_{ijt}x_{ijkt}^{\text{ecv}} + q_{kt}(1 - x_{ijkt}^{\text{ecv}})(i, j) \in \delta(\mathcal{V} \setminus \mathcal{D}^-), k \in \mathcal{K}, t \in \mathcal{T} \quad (2.12)$$

$$0 \leq v_{it} + w_{it} \leq q_{kt} + Q(1 - x_{ijkt}^{\text{ecv}})(i, j) \in \delta^-(\mathcal{V} \setminus \mathcal{D}^+), k \in \mathcal{K}, t \in \mathcal{T} \quad (2.13)$$

$$q_{kt+1} \leq q_{kt} - Qd + Q \sum_{t' \in \{1, \dots, t-1\}} (1 - z_{kt'}^{\text{ecv}})k \in \mathcal{K}, t \in \mathcal{T} \setminus \{n\} \quad (2.14)$$

$$0 \leq q_{kt} \leq Qk \in \mathcal{K}, t \in \mathcal{T} \quad (2.15)$$

$$w_{it} \leq Qy_{it}i \in \mathcal{V} \setminus \{\mathcal{D}^- \cup \mathcal{D}^+\}, k \in \mathcal{K}, t \in \mathcal{T} \quad (2.16)$$

$$y_{it} \geq y_{jt}i \in \mathcal{V} \setminus \mathcal{F}', j \in \mathcal{F}', t \in \mathcal{T} \quad (2.17)$$

$$\sum_{(i,j) \in \delta^-(\{i\})} x_{ijkt}^{\text{ecv}} + x_{ijkt}^{\text{icev}} \leq 1i \in \mathcal{D}^-, k \in \mathcal{K}, t \in \mathcal{T} \quad (2.18)$$

$$\sum_{t' \in \{1, \dots, t\}} z_{kt'}^{\text{ecv}} \geq \sum_{(i,j) \in \delta^-(\{i\})} x_{ijkt}^{\text{ecv}}i \in \mathcal{D}^-, k \in \mathcal{K}, t \in \mathcal{T} \quad (2.19)$$

$$1 - \sum_{t' \in \{1, \dots, t\}} z_{kt'}^{\text{ecv}} \geq \sum_{(i,j) \in \delta^-(\{i\})} x_{ijkt}^{\text{icev}}i \in \mathcal{D}^-, k \in \mathcal{K}, t \in \mathcal{T} \quad (2.20)$$

$$z_{k1}^{\text{ecv}} + z_{k1}^{\text{icev}} = 1k \in \mathcal{K} \quad (2.21)$$

$$z_{kt}^{\text{ecv}}, z_{kt}^{\text{icev}}, y_{it}, x_{ijkt}^{\text{ecv}}, x_{ijkt}^{\text{icev}} \in \{0, 1\}(i, j) \in \mathcal{A}, k \in \mathcal{K}, t \in \mathcal{T}. \quad (2.22)$$

Constraints (2.3) force every customer to be visited exactly once. This single assignment is relaxed in (2.4) for non-customer vertices. To create connected tours, (2.5) impose flow conservation for ECVs, and (2.6) holds for ICEVs. Freight constraints are given by (2.7) and (2.8). Constraints (2.7) impose the delivery freight balance, while Constraints (2.8) ensure that the maximum vehicle freight load is not exceeded at any vertex. Combined with (2.5) and (2.6) they prevent subtours. Constraints (2.9)–(2.11) enforce time window feasibility. Constraints (2.9) determine the start time of servicing or recharging at vertex j after departing from customer vertex i , taking service and driving time into consideration for both ECVs and ICEVs. Constraints (2.10) integrate charging times for ECVs. Constraints (2.11) force start times of servicing or recharging to match the time windows although a vehicle may arrive earlier.

Constraints (2.12)–(2.15) relate to energy consumption and battery degradation. Constraints (2.12) model the energy balance, accounting for energy consumption and recharging processes. Note that although vehicles can start with a fully charged battery, w_{it} is defined for the depot vertex 0 for ease of notation. Since $w_{it} = 0$ is allowed, (2.12) does not yield any loss of generality. Constraints (2.13) limit the vehicle's state of charge to the battery capacity, and constraints (2.14) and (2.15) account for battery degradation once an ECV is purchased. Constraints (2.16)–(2.17) integrate charging station location decisions. While (2.16) only allow charging at vertex i if a charging station is installed at i , constraints (2.17) translate charging station location decisions from real to dummy vertices. Constraints (2.18) ensure that each vehicle covers only one tour per work shift. Constraints (2.19) secure that a sufficient number of ECVs are purchased to cover all tours operated by ECVs. Constraints (2.20) state that once an ECV is bought it must be used. To account for the fleet transformation, constraints (2.21) ensure that an initial investment decision is made for every vehicle. Constraints (2.22) define the domains of the variables.

Some comments on our modeling approach are in order. First, we consider the fleet size to be constant for comparison reasons. This imposes a worst-case bound for ECVs as one cannot compensate lower driving ranges by additional vehicles. In discussions with practitioners, we identified this constraint as a prerequisite to increase the acceptance for ECVs. However, it is straightforward to relax this assumption if it is not needed. Second, we assume that each vehicle is replaced at most once over the time horizon $\mathcal{T}$, which is a plausible assumption since we analyze the diffusion of a new technology whose characteristics may change over a longer time horizon. The LSP's investment decisions focus only on the next replacement as the LSP aims to account for future price and technology developments in succeeding decisions. Third, we include fixed costs and salvage values directly in the investment costs. This does not affect the validity of our model as long as a vehicle is replaced at most once. For settings with multiple replacements, the model can be easily adjusted. Note that we treat driver costs within the operational cost function. Fourth, by imposing (2.21) we assume that every vehicle is replaced once at the beginning of the planning horizon. Although replacing a complete fleet at once appears unrealistic in practice, this assumption is necessary to account for an unbiased comparison between ICEVs and ECVs from a greenfield perspective. However, for specific application cases, (2.21) can easily be relaxed or adapted to consider different settings. Fifth, we assume a linear recharging rate r [kWh/min] when recharging an ECV. Since we consider partial recharging on routes, which is basically linear, this assumption is not limiting. Also, linearizing the recharging curve yields a worst-case estimate, given its concave shape. Sixth, battery degradation depends on complex time- and usage-dependent chemical processes (Pelletier et al., 2017). However, data from field experiments allow us to estimate this degradation with a linear factor (cf. Section 4.2). Again, this yields a worst-case estimate as the shape of the degradation curve is also concave.

3. Methodology

Our planning problem as outlined in Section 2.2 belongs to the class of LRPIFs which are NP-hard. Additionally, compared with conventional LRPIFs for which efficient metaheuristic frameworks are known (e.g., Schiffer & Walther, 2018a), several additional algorithmic difficulties arise. First, our planning problem is made up of a multitude of interdependent planning tasks, namely investment decisions on the fleet composition and on the number and location of charging stations, as well as operational decisions on route plans. Second, we consider a mixed fleet and additional constraints on battery degradation. Third, the investment decisions

are interdependent over time and result in heterogeneous ECVs in terms of battery capacity and battery degradation.

Specifically, the interdependencies between these planning tasks are as follows: an immediate interdependency exists between charging station infrastructure investments and fleet operations for electric vehicles as the location of charging stations impacts feasible routing operations and vice versa, the need for charging stations at certain locations depends on the routing decisions. Once a charging station investment has been made, it also affects operational routing decisions of subsequent time periods. At an upper level, strategic vehicle investment decisions impact the charging station investment and vehicle routing decisions. Vice versa the corresponding charging station investment and vehicle routing decisions determine the economic viability of potential vehicle investment decisions. Again, decisions once taken impact subsequent planning periods.

Considering these difficulties, we develop a novel matheuristic that optimally solves strategic investment decisions (i.e., vehicle and charging station investments) and leaves only operational subproblems (i.e., route plans) to a metaheuristic component, which serves as an oracle to approximate operational costs. In the following, we describe the central solution scheme (Section 3.1) which, for the first time, allows to efficiently branch and prune on subproblems over the time dimension $\mathcal{T}$ for this problem class. After deriving this solution scheme, we elaborate the structure and central components of our algorithm (Section 3.2). We shift detailed explanations of single algorithmic components to Appendix A and Appendix B to keep this paper concise and focused on its managerial dimension. In the same vein, we discard pure numerical studies to proof the solution quality of the metaheuristic component and refer to Schiffer and Walther (2018a) for an extensive numerical evidence.

3.1. Solution scheme

To derive our solution scheme for the time dimension of the MIP defined by (2.2)–(2.22), we discuss some general characteristics of a solution. We refer to a solution $\sigma = (\mathbf{z}^{\text{ecv}}, \mathbf{y}, \mathbf{x})$ as a triplet made up of an ECV investment vector $\mathbf{z}^{\text{ecv}}$, a charging station vector $\mathbf{y}$, and an arc vector $\mathbf{x}$. The variable $\mathbf{z}^{\text{ecv}}$ contains decisions on all z_{kt}^{ecv} , and $\mathbf{y}$ on all y_{it} , while any $x_{ijkt}^{\text{ecv}}, x_{ijkt}^{\text{icv}} \in \mathbf{x}$ contains information on the traversed arcs (i, j) as well as on the traversing vehicle k and the time period t . Since $\mathbf{z}^{\text{icv}}$ results from $\mathbf{z}^{\text{ecv}}$, it is not necessary to include it in σ . Note that all variables stored in σ are binary.

The main idea of our solution scheme is to efficiently explore one vector in σ over time period t by branching and pruning, and solving the resulting subproblems efficiently with a metaheuristic. While the number of subproblems is crucial for the computational complexity of our algorithm, as it heavily affects the solution space of the branch-and-prune component, the size of the subproblems is less critical as these are solved heuristically. Hence, we choose $\mathbf{z}^{\text{ecv}}$ as our guiding vector according to Proposition 1.

Proposition 1. Choosing $\mathbf{z}^{\text{ecv}}$ as the guiding vector in the MIP always yields the lowest number of subproblems for (2.2)–(2.22).

Proof. The number of subproblems resulting from a binary vector $\mathbf{v}$ is given by the cardinality of its permutation set $\Pi(\mathbf{v})$. Then

$$\begin{aligned} |\Pi(\mathbf{z}^{\text{ecv}})| &= (|\mathcal{K}||\mathcal{T}|)! << |\Pi(\mathbf{y})| \\ &= (|\mathcal{V}||\mathcal{T}|)! << |\Pi(\mathbf{x})| = (|\mathcal{A}||\mathcal{K}||\mathcal{T}|)! \end{aligned}$$

holds as long as

$$|\mathcal{K}| < |\mathcal{V}| < |\mathcal{A}|.$$

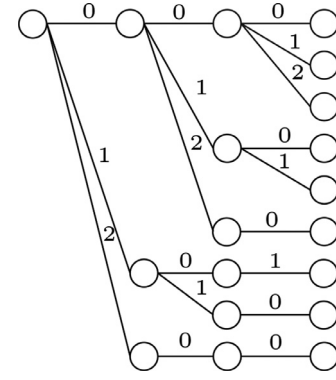


Fig. 1. Example of a vehicle investment decision tree Ψ for $|\mathcal{K}| = 2$ and $\mathcal{T} = \{0, 1, 2\}$.

Even in the synthetic case of $|\mathcal{K}| = |\mathcal{C}|$ and $\mathcal{F} = \emptyset$, with $|\mathcal{K}| = |\mathcal{V}| < |\mathcal{A}|$,

$$\begin{aligned} |\Pi(\mathbf{z}^{\text{ecv}})| &= (|\mathcal{K}||\mathcal{T}|)! \\ &= |\Pi(\mathbf{y})| = (|\mathcal{V}||\mathcal{T}|)! < |\Pi(\mathbf{x})| = (|\mathcal{A}||\mathcal{K}||\mathcal{T}|)! \end{aligned}$$

remains, so that guiding $\mathbf{z}^{\text{ecv}}$ still yields the lowest number of subproblems.

This concludes the proof. $\square$

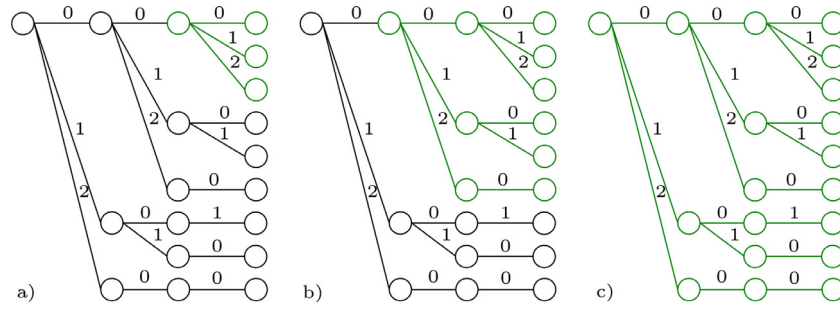
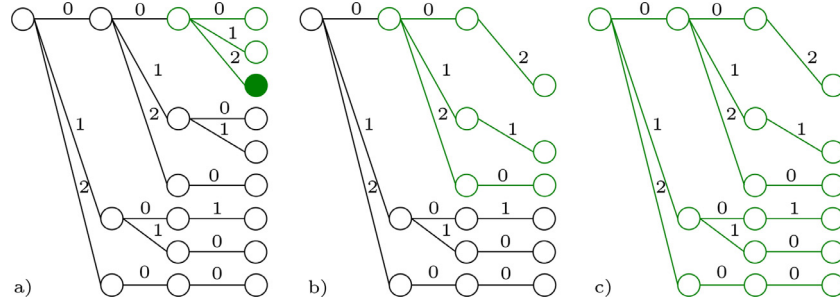
We formally represent a tree Ψ as a set of its paths rooted at the origin and ending at a terminal node. Fig. 1 depicts a decision tree Ψ for vehicle investments if $|\mathcal{K}| = 2$ and $\mathcal{T} = \{0, 1, 2\}$. The number on each arc indicates how many ECVs are purchased in the respective period. Solving this problem straightforwardly implies evaluating each path of the decision tree separately. For each path $\psi \in \Psi$, a clustered LRPIF with a mixed and heterogeneous fleet must be solved. Clusters are used to represent different time steps $t \in \mathcal{T}$ (cf. Appendix A). The solution for each path in Ψ yields the route plans for each time step, but also time-interdependent investment and location decisions on charging stations. Furthermore, investment decisions at time t influence the evaluation and decisions in all consecutive time periods. Hence, a straightforward evaluation scheme requires a complete evaluation of each path ψ over $\mathcal{T}$ with $\Theta^{fw} = (|\mathcal{K}| + 1)|\mathcal{T}| = |\Psi|$ computational runs, each evaluating an instance with a size of $\Lambda^{fw} = |\mathcal{V}||\mathcal{T}|$. Such an approach may prove computationally tractable for some artificial cases, but shows unacceptable high computational times and effort for most real-world planning tasks.

We next prove Proposition 2 to reduce the complexity of our guiding strategy.

Proposition 2. For a subproblem of (2.2)–(2.22) with $\mathcal{T}' \subset \mathcal{T}$: $\mathcal{T}' \nmid t' \in \mathcal{T}', t' < t \in \mathcal{T} \setminus \mathcal{T}'$ the number of ECVs of the optimal solution of the subproblem yields a lower bound on the number of ECVs of the optimal solution of (2.2)–(2.22).

Proof. If we expand $\mathcal{T}'$ for additional time periods of $\mathcal{T}$, three possibilities remain: (i) ECVs are not purchased in earlier periods because this does not yield additional savings; in this case the lower bound is tight; (ii) (some of) the ECVs should be acquired at an earlier time period because this yields additional savings; in this case the lower bound also remains tight; (iii) by increasing $|\mathcal{T}'|$, the number of ECVs also increases due to additional savings; in this case, the lower bound further increases and may remain tight or further increase in the next extension of $\mathcal{T}'$. $\square$

Proposition 2 implies that instead of computing (2.2)–(2.22) forward in time, evaluating subproblems backward is possible and allows for additional pruning due to lower bounds on the number of ECVs (instead of exploring the complete space of subproblems).

Fig. 2. Backwards search strategy for $|K| = 2$ and $T = \{0, 1, 2\}$.Fig. 3. Reduced backward search tree with pruning due to $LB = 2$.

```

1:  $T' \leftarrow \{t^{end}\}$ 
2: while  $|T'| \leq |T|$  do
3:    $\iota \leftarrow 0$ , initializeParameters()
4:   while isActive(vehicleInvestment()) do
5:      $z^{ecv} \leftarrow \text{adjustVehicleInvestment}()$ ,  $\sigma' \leftarrow \text{InitialSolution}()$ 
6:     while ( $\iota \leq \iota^{max}$ ) do
7:       if  $\iota \bmod \iota^{opt}$  then
8:         optInvest()
9:         solveSubproblem()
10:       $\iota \leftarrow \iota + 1$ 
11:   enlarge( $T'$ )

```

Fig. 4. Pseudocode overview of the matheuristic.

This is the core idea of our search strategy that helps reduce the computational complexity significantly.

Figs. 2 and 3 depict an example of this search strategy. We first solve the green colored subproblem with $T' = \{t^{end}\}$ (2a). Then we consecutively extend T' for its closest preceding time step (2b), until we end up solving the complete problem (2c). With this strategy, we reduce the number of computations to $\Theta^{bw} = |T| < \Theta^{fw}$, which significantly reduces the number of necessary computations compared with a forward evaluation. Additionally, we reduce the size of the instances at each iteration to $\Lambda^{bw} = |\mathcal{V}| |T'| \leq \Lambda^{fw} \quad \forall T'$. By so doing, we need to solve a problem having the worst-case complexity only once, in the last iteration of our algorithm. Following this evaluation scheme, we no longer evaluate a complete path $\psi \in \Psi$, but we are capable of successively making the investment decisions for paths of our subproblem. Fig. 3 illustrates how the complexity may be even further reduced due to instance-specific pruning on the lower bound of the number of ECVs.

3.2. Algorithm

Fig. 4 provides the pseudocode of our algorithm. Our solution scheme allows to solve subproblems of (2.2)–(2.22) with respect to the time dimension and the vehicle investment decision by branching and pruning successively: we iteratively increase the number of time steps considered in our subproblem, and also shift the vehicle investment decision (*adjustVehicleInvestment*()) to our

solution scheme as discussed in the preceding paragraph. While for the first subproblem ($T' = \{t^{end}\}$) we have to consider all possible fleet compositions, i.e., vehicle investments, we use a lower bound on ECV investments in the succeeding subproblems to reduce the size of the decision tree in the remaining subproblems (cf. Fig. 3).

We solve each subproblem and vehicle configuration, i.e., a partial subproblem for ι^{max} iterations. We split the remaining decisions into two algorithmic components: at each iteration ι , the *solveSubproblem*() routine solves the operational decision on the route plans and recharging stops (if necessary) with an efficient metaheuristic. Every ι^{opt} iterations, the *optInvest*() routine (re-)solves the remaining investment decisions on charging station investments and locations in order to account for the interdependencies between the operational and strategic decisions. Once a subproblem is solved, we enlarge T' for the next time step and solve the resulting subproblem until $T' = T$. To keep this paper concise, we refer to Appendix A for details on our state of the art metaheuristic and explain only the novel investment decision procedure in the following.

The charging station investment decision depends heavily on the quality of the route plans that remain a heuristic input. Hence, we first use a set partitioning algorithm to improve these route plans. Doing so, we also take advantage of already exploited regions of the solution space between subproblems. We use a solution pool ϑ (see Appendix A) and an index set $\mathcal{R}$ for all routes in ϑ that consists of an index set for ECV routes $\mathcal{R}^{ecv}$ and an index set for ICEV routes $\mathcal{R}^{icev}$. We use additional subsets $\mathcal{R}_i \subseteq \mathcal{R}$ for the set of routes that contain a customer $i \in \mathcal{C}$, and subsets $\mathcal{C}_t \subseteq \mathcal{C}$ for the set of customers that belong to cluster $t \in \mathcal{T}$. Let c_i be the cost of a route i , and let x_{it} be a binary variable indicating whether route i is chosen in cluster t ($x_{it} = 1$) or not ($x_{it} = 0$). Furthermore, Z_t^{ECV} and Z_t^{ICEV} denote the number of ECVs and ICEVs used in cluster t . With this notation, we derive our set partitioning model as follows. In addition to a conventional set partitioning, we ensure that the right number of ECV (3.3) and ICEV (3.4) routes are chosen for any cluster:

$$\text{minimize} \sum_{i \in \mathcal{R}, t \in \mathcal{T}} c_i x_{it} \quad (3.1)$$

subject to

$$\sum_{i \in \mathcal{R}_j} x_{it} = 1 \quad j \in \mathcal{C}_t, \quad t \in \mathcal{T} \quad (3.2)$$

$$\sum_{i \in \mathcal{R}^{\text{ECV}}} x_{it} = Z_t^{\text{ECV}} \quad t \in \mathcal{T} \quad (3.3)$$

$$\sum_{i \in \mathcal{R}^{\text{ICEV}}} x_{it} = Z_t^{\text{ICEV}} \quad t \in \mathcal{T} \quad (3.4)$$

$$x_{jt} \in \{0, 1\} \quad j \in \mathcal{C}_t, \quad t \in \mathcal{T}. \quad (3.5)$$

Instead of solving this problem with a standard mixed integer program solver, we use a depth-first backtracking algorithm and additional lower bounds to yield fast computational times. If the route plan found by our set partitioning algorithm improves upon the route plan of the current solution, we replace the current solution. Based on this improved solution, we then take the charging station investment decision as follows:

- (i) First, we solve a second set partitioning problem for each cluster by assigning each ECV to a route, such that the total battery capacity violation is minimized.
- (ii) Second, we optimally insert charging stations to remove battery constraint violations if necessary. We use a dynamic programming routine that optimally determines visits to charging stations using the resource extension functions introduced in Schiffer and Walther (2018a).

This allows us to keep the number $\iota^{\max}$ of iterations per subproblem low without worsening the solution quality, which helps derive a computationally tractable algorithm. Once a subproblem is solved, we enlarge $\mathcal{T}'$ for the next time step and solve the resulting subproblem until $\mathcal{T}' = \mathcal{T}$.

4. Experimental design

Our aim is to provide a generic analysis of the deployment of ECVs in mid-haul logistics to evaluate the operation and competitiveness of ECVs to derive viable transformation strategies for logistics fleets. A real-world case from a German retail company, the TEDi GmbH, provides the basis for our analyses and ensures practical relevance. To derive general managerial insights, we enrich this basic case study with an in-depth sensitivity analysis on network structures and technology developments. In the following, we first detail the fundamentals of the TEDi case in Section 4.1, before performing our sensitivity analyses in Section 4.2.

4.1. The TEDi case - network structure and general data

We base our analyses on the real-world logistics network of the German retail company TEDi, which operates about 1400 stores all over Europe, selling a broad range of non-food items. Here we focus on a representative distribution area of the overall network. This area (cf. Fig. 5) is located in North Rhine-Westphalia, a federal state of Germany. In this area, 302 stores are supplied from a central warehouse of about 42,000 square meters, located in Dortmund. The stores are scattered in a vicinity of 190 kilometers from the central warehouse. TEDi currently operates a fleet of medium-duty ICEVs, i.e., 12-tonne trucks that can carry 18 pallets with a payload of about five tonnes. Each truck is driven during two shifts per day, i.e., early morning and afternoon shifts. Data on average weekly demand of each store for the year 2015 are available. The median delivery size is six pallets, but it varies significantly in between stores (standard deviation $\sigma = 2.06$, min = 4, max = 12), while it remains stable for each single store. We consider the data of this network to be representative of a mid-haul retail logistics

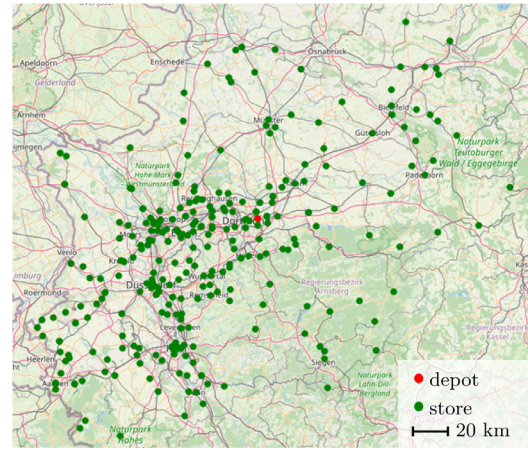


Fig. 5. Logistics network structure of TEDi in Northrhine Westfalia.¹

network with respect to the spatial distribution, the number of customers, and the supply structure of the network.

To study fleet transformation and investment decisions for this case study, we aim at a worst-case scenario from an ECV perspective in order to avoid a positive bias towards ECVs. Accordingly, we focus our analysis on the smallest plausible time horizon because any additional time period would work in favor for ECVs as their lower operational cost would gain impact in the overall TCO analysis. We assume that an operator keeps a vehicle at least until the end of its economic lifetime from a tax law perspective. Thus, we set our planning horizon $\mathcal{T}$ to five years, considering a linear depreciation of 20% per annum according to German tax law (BMF, 1998). To calculate TCO we consider an annual discount rate of 5% (Taefi et al., 2017). We use a driver wage of 29,000 € per year (Destatis, 2017). Further, we neglect vehicle taxes that do not differ between ECVs and ICEVs and we account for all cost values without VAT or other refundable taxes. To convert monetary values from dollar to euro, we use a conversion factor of 0.8122 € / \$.

4.2. Sensitivity analyses

Although we base our study on a real-world logistics network, several uncertainties require in-depth sensitivity analyses in order to expand the findings of a basic case to general managerial insights, since ECVs are an emerging technology for which significant price and technology uncertainties exist for both vehicles and charging stations. We derive a technology roadmap (4.2.1) to account for these uncertainties. Second, uncertainties exist with regard to emission savings as these savings significantly depend on the electricity grid mix which is used to recharge ECVs. Hence, we account for different electricity mix scenarios (4.2.2). Third, although the TEDi network is representative of a mid-haul logistics network from a spatial perspective, network structures may change with respect to temporal customer patterns and logistics operations. To take these effects into account, we derive additional network scenarios (4.2.3).

4.2.1. Technology roadmap

Information on prices and characteristics of mid-haul ECVs, batteries, and charging stations is still scarce, and forecasts on future developments are fraught with high uncertainties. Existing studies usually consider a single application case for a static

¹ The figure bases on OpenStreetMap, which is data licensed under the Open Data Commons Open Database License (ODbL) by the OpenStreetMap Foundation (OSMF). For more information see www.openstreetmap.org/copyright.

point in time and hence only apply unsound sensitivity analyses on technology and cost parameters if any. Such time-invariant sensitivity analyses may lead to unrealistic combinations of price ratios and technologies. To remedy these drawbacks, we derive settings from a technology roadmap based on reliable company information, field experiments, market forecasts, and governmental studies. We investigate the development of technological and cost parameters for (i) vehicles, (ii) batteries, and (iii) charging stations for the 2015–2030 time span.

ICEV data: We focus on 12.5-tonne trucks, which is a standard truck size for mid-haul logistics and is also the truck size that is used by TEDI. We derive costs and vehicle characteristics from an ICEV that is currently offered on the market, the MAN TGL 12.250. We assume an average investment costs of 75,000 € (Taefi et al., 2017). To calculate operational costs, we consider a fuel consumption rate of 0.19 L/kilometre, a fuel price of 0.96 € /L based on the average German fuel price of 2015 (Union, 2016b), and maintenance costs of 0.26 € /kilometre (Davis & Figliozzi, 2013). Since ICEV vehicles are a mature technology, we do not consider any technological developments or price changes over time.

ECV data: Currently, information on mid-haul ECVs, i.e., 12.5-tonne electric trucks, is still sparse as only few pioneering companies offer models on the market. Since 2015, the company EMOSS has been selling 12.5-tonne trucks with a battery capacity of 160 kilowatt-hours (i.e., a range of 200 kilometre) for 160,000 € (EMOSS, 2016). Regarding future developments, Tesla has announced the market launch of the Tesla Semi truck models with a range of 480 kilometre (Semi300) and 800 kilometre (Semi500) and market prices of 122,000 € and 147,000 €, respectively, for the year 2019 (Tesla, 2018). Since we want to carry out a rather conservative study avoiding overestimates in favour of ECVs, we assume a market delay of two years regarding the launch of the Semi300 and an additional two year shift between the two vehicle types, i.e., thus considering a market launch in 2021 for the Semi300 and in 2023 for the Semi500. Additionally, we reduce the estimated range by 20%.

Currently, information on real-world operational performance and costs of ECVs is very scarce. However, a fleet of mid-haul ECVs was operated in real-world settings within the field project ELMO (Electrified Commercial Transport in Urban Areas) which was initiated and partly funded by the German government (cf. Stütz et al., 2016). Among other companies, TEDI participated in this project and used two 12-tonne medium-duty electric trucks from EMOSS (model CM1216) (EMOSS, 2016) in daily operations within a catchment area of 70 km around the central warehouse. As more than 3000 transportation records were gathered covering a total mileage of 158,209 km and an energy consumption of 108,543 kilowatt-hours within ELMO, this provides a unique opportunity to assess the viability of ECVs in mid-haul logistics. Based on these data, we derive information on real-world travel times, energy consumption and recharging rates. In order to calculate resulting operational costs, we assume the average energy price for a company that consumed between 2000 and 20,000 megawatt-hours per year in 2015, resulting in 0.1308 €/kilowatt-hour (Union, 2016a). Additionally, we apply distance-dependent maintenance costs of 0.13 €/kilometre (Davis & Figliozzi, 2013).

Battery data: To model price decreases for batteries, we estimate a learning curve using a shape-preserving piecewise cubic interpolation. We consider the average market prices of the years 2010 to 2016 (cf. BNEF, 2017), a price estimate of 125 €/kilowatt-hour for the year 2022 (DoE, 2017), and an estimate of 74 €/kilowatt-hour for the year 2030 (BNEF, 2017). Current prices stated for 2017 and 2018 by different companies are consistent with our approximations (Holland, 2018).

Battery degradation, i.e., a decrease in the capacity of lithium-ion batteries over the batteries' lifetime, results from calendar and

Table 3
Different scenario characteristics.

Vehicle technology	Charging station type
eMoss / Semi300 / Semi500	50 kilowatt (p_1 – p_6), 150 kilowatt (p_7)

cycle aging (Pelletier et al., 2017). Field experiments have shown that batteries have a residual capacity of at least 80% after 3000 full charging cycles (Davis & Figliozzi, 2013). Thus, a capacity of at least 80% remains after a five-year utilization period, assuming 260 working days per year and a conservative maximum of two full charging cycles per day. We account for a degradation of 4% per year, which remains a worst-case estimation given the concave shape of the degradation curve.

Charging station data: In mid-haul logistics, it is necessary to recharge vehicles en route. Hence we assume that fast charging stations have to be installed. For the 50 kilowatt fast charging stations that currently exist, set up costs can differ significantly between 3320 € and 42,330 €, depending on the location and network topologies as observed in field projects (DoE, 2015). We assume that 50 kilowatt fast chargers are available since 2015 in the [4000€, 44000€] price range. Regarding future developments, companies are testing and launching first 150 kilowatt charging stations (ABB, 2017). We also assume that 150 kilowatt fast charging stations are available from 2021 at the upper price limit of the current 50 kilowatt charging stations, which is in line with governmental plans and forecasts (cf. NPE, 2014). We consider annual maintenance costs of 243.66 € and a linear depreciation over seven years (Snyder et al., 2012).

Fig. 6 details this roadmap with respect to vehicle characteristics, battery prices, and charging technologies. It shows the price forecast for batteries from 2015 to 2030. Furthermore, it highlights the availability of new charging technologies and new vehicle classes over this time horizon. Based on this technology roadmap, we identify the following characteristics to differentiate the scenarios for our sensitivity analysis as shown in Table 3.

We first create separate scenarios for the availability of each vehicle type (eMoss, Semi300, Semi500). Analyzing the battery price development in our technology roadmap (Fig. 6a) and taking the availability of the different vehicle types into account (Fig. 6b), one can see that the price differences in the vehicle types already reflect the development in battery prices. Since battery prices are expected to reach a lower threshold after 2023, we do not consider further price decreases in vehicle costs by battery price developments to keep our scenarios as conservative as possible.

We further differentiate these scenarios with respect to the available charging station technology and its price. With a price step width of 8000 €, we derive six 50 kilowatt charger scenarios with increasing prices in the range of [4000€, 44,000€] ($[p_1, p_6]$), and one 150 kilowatt charger scenario with a price of 44,000 € (p_7).

4.2.2. Emissions

To calculate CO₂ emissions for ICEVs, we use the CO₂ data for diesel from Juhlich (2016). With an average fuel demand of 0.19 l/kilometre this amounts to 716 gCO_{2e}/kilometre. For ECVs, the resulting CO₂ emissions vary heavily depending on the electricity mix used to charge the vehicles. We consider four different electricity mix scenarios based on data from the German environmental protection agency, accessible at <http://www.probas.umweltbundesamt.de/>. First, we calculate the CO₂ emissions for ECVs based on the average German grid mix in 2020 (mix20) with 288 gCO_{2e}/kilometre and in 2030 (mix30) with 187 gCO_{2e}/kilometre. Additionally, we consider the emissions for an altered grid mix, assuming that additional electricity demand

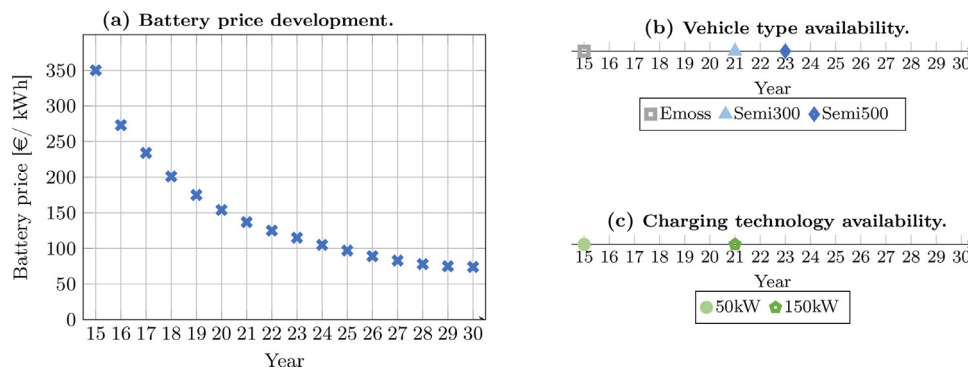


Fig. 6. Technology roadmap.

Table 4
Different scenario characteristics.

Vehicle technology	Charging station type	Daily shifts	Customer demand	Demand quantity
eMoss / Semi300 / Semi500	50 kilowatt (p_1 – p_6), 150 kilowatt (p_6)	1 / 2	static / varying	full / halved

for electric vehicles is covered by renewables in 2030 (renew) with 158 gCO_{2e}/kilometre. Finally, we consider a grid mix in 2030 which is largely based on renewable energies (15% wind onshore, 55% wind offshore, 20% water), and contains only 10% of gas-powered plants to balance the volatility of the renewables (renew) with 31 gCO_{2e}/km.

4.2.3. Network characteristics

Retail logistics networks exhibit a wide variety of characteristics, e.g., the TEDi network is characterized by (i) static customer patterns over a long period of time and (ii) route plans constrained by the trucks freight volume (and not by the maximum load) as lightweight goods are transported on pallets. To account for a variety of network characteristics, we apply the following modifications:

- We create scenarios in which customer patterns vary for each time step. In these scenarios, we shuffle the original demand distribution among all customers for each time step.
- To account for weight-constrained instead of volume-constrained route plans, we create scenarios in which the original demand is halved.
- The general utilization of vehicles, influenced by the number of daily driven shifts, may heavily influence our results. Hence, we generate additional scenarios with a single daily shift.

Taking these modifications into account, we obtain the scenario characteristics shown in Table 4. In our studies, we apply a full-factorial design for these characteristics, resulting in 168 scenarios.

5. Results

For our studies, we compute two baselines for each scenario. We first compute a solution for a pure ICEV fleet (ICEV-only). Assuming that a fleet operator is not willing to increase her number of vehicles or drivers within a transformation process, this solution denotes the fleet size for our remaining studies. We then compute a solution for a pure ECV fleet (ECV-only) over the entire time horizon, which allows us to evaluate the maximum ecological benefit in terms of emissions savings and to compare an abrupt complete fleet transformation with a smooth and potentially partial transition over time. We compare these two baselines to the cost-optimal solution from our planning approach, which allows for vehicle and charging station investments over time. In the following we synthesize the main findings from these results.

First, we detail results for our real-world case study of the TEDi GmbH in Section 5.1. We then generalize these findings for all 168 scenarios in Section 5.2.

To keep this paper concise, we present a detailed summary of the results for all scenarios with respect to overall costs, network design, and operational quantities in the supplemental material of this paper.

5.1. Results of the TEDi Case

Fig. 7 details the operational characteristics (7)a and the corresponding cost structure (7)b for the ICEV-only, ECV-only, and cost-optimal (CO) solution of the TEDi network. From Fig. 7a it can be seen that although the number of vehicles remains equal for all solutions, both the ECV-only and the cost-optimal solution show slightly increased driver shifts and total driven distances. Although the number of driver shifts is the highest for the cost-optimal solution, the driven distance is lower than for the ECV-only solution as a mix of ICEVs and ECVs allows for a more efficient assignment of tours to vehicles.

Fig. 7b shows the total cost for each solution, and the respective cost shares for vehicle investment (C^{veh}), driver (C^{driver}), operational (C^{oper}), and charging station investment (C^{stat}) cost. As can be seen, even an ECV-only strategy can compete with an ICEV fleet, as it yields a cost reduction of 0.61% (24,214 €). Deploying a cost-optimal mixed fleet yields a higher cost reduction of 2.81% (111,421 €). This shows that for certain application cases in retail mid-haul logistics, e.g., the TEDi network of North Rhine-Westphalia, ECVs are already breaking even compared with ICEVs. In these cases, reduced operational costs overcompensate larger investment costs for vehicles and charging infrastructure.

To detail the time-dependent nature of investment decisions, Fig. 8 shows the vehicle investments (8)a and charging station investments (8)b over time for the cost-optimal (CO) deployment strategy, as well as the charging station investments (8)c for the ECV-only (EO) strategy. Although ECVs could be deployed over the whole time horizon in the cost-optimal case, most vehicles are purchased at the beginning to leverage the full potential of their operational cost savings. Only a single vehicle is purchased at time $t = 2$ to allow to better shift vehicles with degraded batteries between routes, which allows to reduce the number of necessary charging stations from 57 to 52. For charging station investments, both the ECV-only and the cost-optimal solution reveal similar

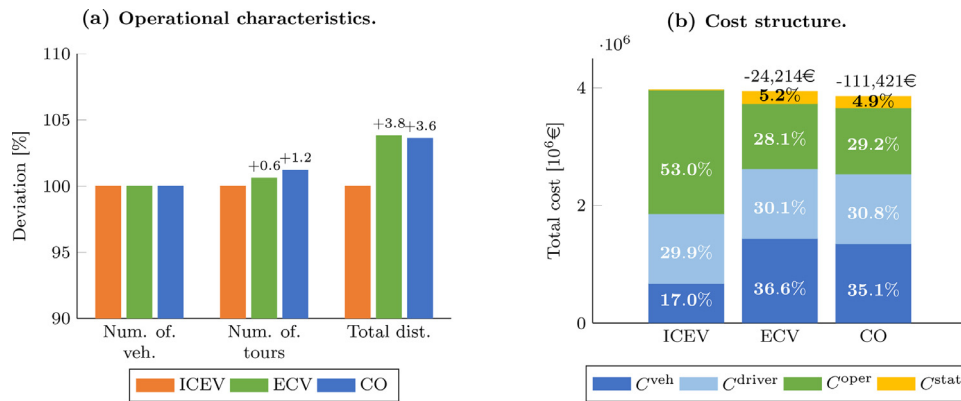


Fig. 7. Operational characteristics and cost structure for the ICEV-only, ECV-only, and cost-optimal solution of the TEDi network.

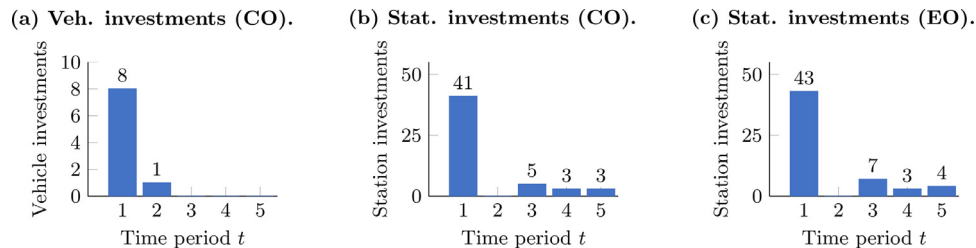


Fig. 8. Investment decisions over time for vehicles and stations in the cost-optimal (CO) and stations in the ECV-only (EO) solution for the TEDi case.

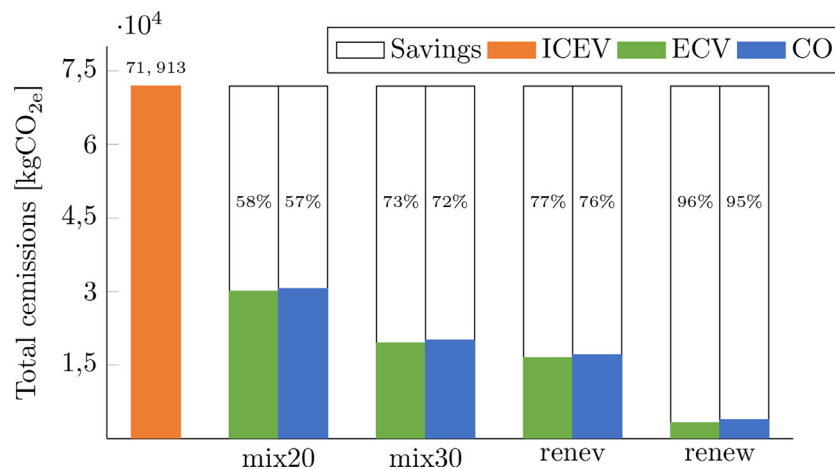


Fig. 9. Average emission savings for the ECV-only solutions and the cost-optimal solutions compared with the ICEV solution for the different grid mix scenarios in the TEDi case.

strategies. All charging stations that are necessary to keep the fleet operational are purchased at time $t = 1$, whereas the required investments in charging stations that are due to battery degradation at a later stage to keep vehicles operational, are prolonged as much as possible.

Fig. 9 shows the respective emission savings for the ECV-only and the cost-optimal solution and each electricity mix as defined in Section 4.2.2. With an ECV-only solution, significant emissions savings of at least 58% (41,879 kgCO_2e) and up to 96% (68,680 kgCO_2e) can be reached. Even the cost-optimal solution yields at minimum emission savings of 57% with the worst electricity mix, so that the emissions of the TEDi network are at least halved.

Concluding, the TEDi case represents a very promising deployment scenario in which both the ECV-only and the cost-optimal solution outperform the ICEV-only solution, so that significant emissions savings can be reached in all scenarios. However, these

results may vary significantly, depending on network, vehicle, and charging station characteristics. Hence, we generalize and discuss these findings for all scenarios in the following.

5.2. General results

Our analyses have so far focused on the basic scenario of the TEDi GmbH, which is limited to a certain network structure and the eMoss vehicle type. We now extend these findings to all scenarios, pointing out differences between the vehicle types, charging technologies, and network structures.

Specifically, the narrative of our sensitivity analysis is as follows: regarding driving ranges, we show that scenarios with ECV vehicles with higher driving ranges lead to a significantly higher diffusion of ECVs, independent of other scenario characteristics. Regarding charging station investments, we show that less in-

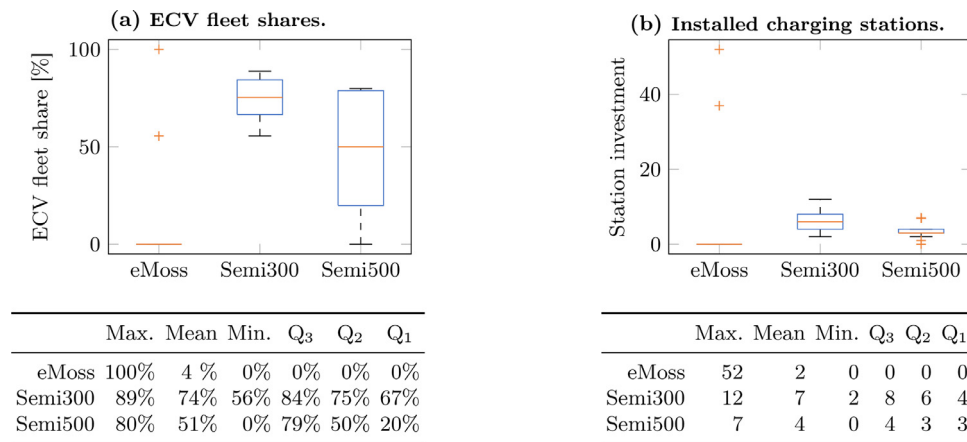


Fig. 10. ECV share and total number of charging station investments over all scenarios for each vehicle type.

Table 5
Number of scenarios with ECV investments.

	eMoss	Semi300	Semi500
Total number of scenarios	56	56	56
Scenarios with ECV investment	3	56	56
Scenario share with ECV investments	5%	100%	100%

investments into charging stations are necessary for vehicles with higher driving ranges, which further supports the diffusion of ECVs. Regarding vehicle investments, we show that the timing of investment decisions also depends on the ECV type. While both Tesla Semi models lead to investment strategies that purchase almost all ECVs at the beginning of the planning period, investment strategies for the eMoss model split investments over two time steps. Finally, we compare a forced electrification of the whole fleet to an optimal fleet transformation with respect to cost and emission savings. We show that a full electrification of a fleet may cause additional total cost of up to 50% compared with a partial cost optimal transformation. In general, emission savings differ with respect to the underlying electricity mix scenario.

Fleet structure: Table 5 shows the numbers and share of scenarios in which the cost-optimal solution comprises ECV investments for the three ECV types (eMoss, Semi300, Semi500). As can be seen, the TEDi case analyzed in Section 5.1 denotes a positive exception for the eMoss scenarios, as only two additional scenarios exist in which ECVs are deployed. In contrast, in the Semi300 and the Semi500 scenarios the fleet is always at least partially electrified.

Charging infrastructure investments: To analyze the scenarios with ECV deployment in more detail, we show in Fig. 10 the ECV fleet share (10)a and the total number of installed charging stations (10)b. As can be seen, the TEDi case is the only eMoss scenario that yields a full electrification, whereas the other two yield an electrification slightly above 50%. Surprisingly, the Semi300 and Semi500 scenarios do not yield a full electrification either, as the different investment costs favor a partial electrification of the fleet. This partial electrification appears to be viable in nearly all scenarios for the Semi300 and Semi500. For both the Semi300 and the Semi500 scenarios, the number of necessary charging stations is significantly lower than in the eMoss scenarios since the driving range is sufficiently large, so that the charging infrastructure is built only at the depot to account for recharging overnight and between shifts.

Vehicle investments: Focusing on the temporal characteristics of the investment decisions related to Fig. 10, we depict in Fig. 11

the share of ECV investments for each vehicle type. As can be seen, for the eMoss solutions, vehicle investments are made in the first and second time periods since in some cases delayed investments decrease the need for additional charging stations. For the Semi300 and Semi500 case in which no recharging en route is necessary, nearly all ECVs are purchased in the first period to maximize operational cost savings. Focusing on charging station investments, the eMoss scenarios show investment patterns as discussed in Section 5.1 with only a few delayed investments and the majority of charging stations being built at the beginning. All charging station investments for the Semi300 and Semi500 scenarios occur at the beginning since the charging infrastructure is only deployed at the depot.

The price of full electrification: Fig. 12 connects the results discussed so far to a cost perspective and shows a box-whisker plot for the cost deviations of the ECV-only (E) and the cost-optimal (CO) solutions compared with the ICEV-only solution for the scenarios of each vehicle type. As can be seen, the eMoss scenarios contain only one economically worthwhile ECV-only solution, whereas additional costs of up to 81.96% may arise in the other scenarios. Two other scenarios suggest a partial electrification that yields cost savings of up to 7.09%. Focusing on the Semi300 and Semi500 scenarios, one can see that the cost-optimal solution always yields cost savings by partially deploying ECVs. However, these savings may reach 15% depending on the scenario characteristics. Focusing on the ECV-only solutions, the Semi300 solutions yield cost savings for all but four scenarios. These are scenarios with 150 kilowatt charging stations at a price of 44,000€ and a single daily shift. In this case, the additional charging station investment costs exceed the operational cost savings. For the Semi500 solutions additional cost increases of up to 10.33% arise in 50% of the scenarios. This shows that for the considered network size, the additional driving range of the Semi500 does not further improve the network operations, so that the total benefit is only decreased by higher vehicle investment costs.

Emission savings: Fig. 13 shows the average emissions savings over all scenarios for the different vehicle types and electricity grid mix scenarios of Section 4.2.2 for the ECV-only solutions (13)a and the cost-optimal solutions (13)b. Fig. 13a validates the emission savings range already analyzed in Section 5.1 for all scenarios. However, these solutions may come with a significant increase in total cost, especially for the eMoss scenarios (cf. Fig. 12). Focusing on the cost-optimal solutions, Fig. 13b illustrates the general potential of emission savings. While the Semi300 scenarios still show an emissions saving potential above 50%, the Semi500 scenarios yield significantly smaller savings potentials compared with the ECV-only solution.

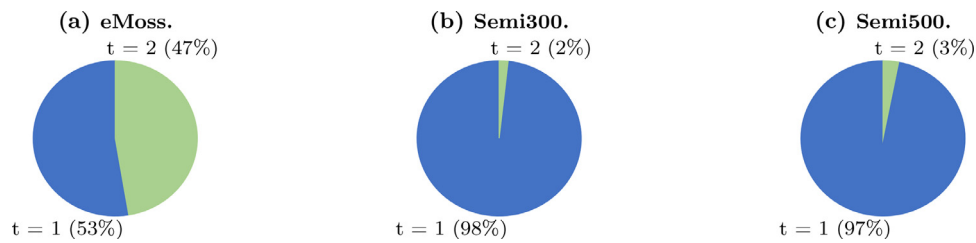


Fig. 11. ECV investment shares for the eMoss, Semi300, and Semi500 scenarios per time period.

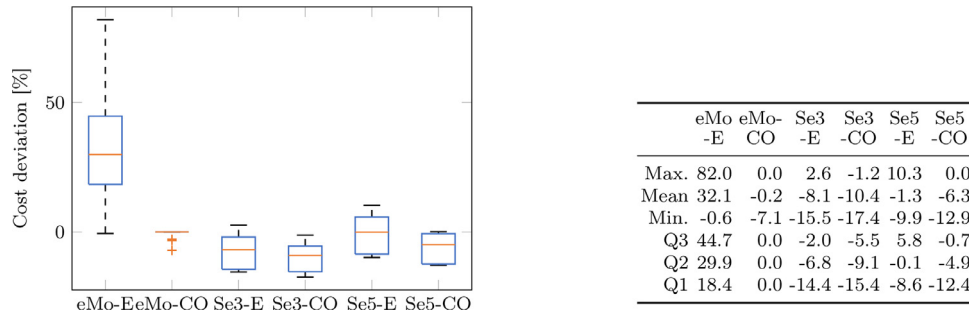


Fig. 12. Total cost deviations for the eMoss (eMo), Semi300 (Se3), and Semi500 (Se5) scenarios between the ECV only (E) and the cost-optimal (CO) solution compared to the ICEV solution.

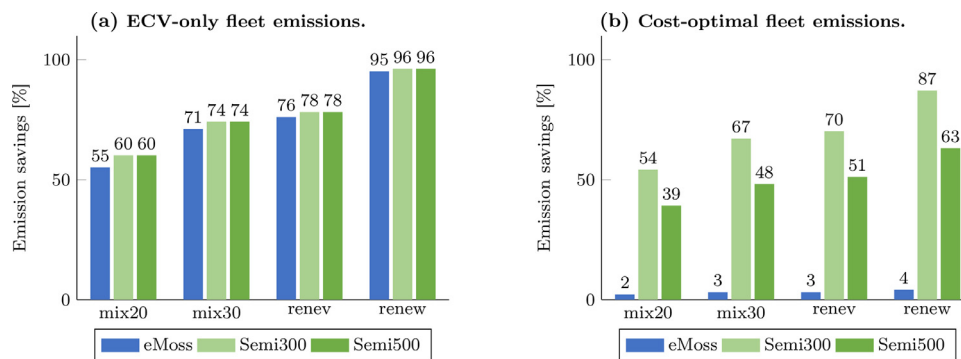


Fig. 13. Average emission savings compared to the ICEV solution for the different vehicle type scenarios.

Focusing on the eMoss scenarios, only the few economically worthwhile scenarios yield emissions savings, so that on average the emissions savings nearly disappear. However, for certain cases the emissions savings as discussed in Section 5.1 remain.

Conclusion: Concluding, the economic and ecological potential and benefits of deploying ECVs in mid-haul logistics networks can hardly be generalized. With current technology settings (i.e., in the eMoss scenarios) only very few network typologies exist for which deploying ECVs yields an economically viable solution. In these network typologies, the solutions further depend very strongly on the total cost of charging infrastructure necessary to operate the network. However, for these few cases, emissions savings of more than 50% can be realized. For upcoming technology settings, a partial electrification of mid-haul logistics networks appears to always be economically worthwhile. Nevertheless, oversizing technological characteristics for vehicles (Semi300) or charging stations (150 kilowatt charger) may decrease the share of ECVs in a mid-haul logistics fleet. In most cases, the cost-optimal solution still remains a partial electrification of the fleet since some routes are always better operated by ICEVs. Focusing on emissions savings, a cost-optimal partial solution for a technological setting in which a vehicle's driving range and charging station infrastructure are not oversized (i.e., the Semi300 scenarios) still yields emissions savings close to those realized with a full fleet electrification. In contrast, oversized technologies (i.e., the Semi500 scenarios)

may significantly decrease the share of deployed ECVs due to their higher cost and hence may significantly reduce the emissions savings.

6. Conclusions

We have developed a novel modeling approach to assess the viability of ECVs in mid-haul logistics networks integrating investment decisions over time and a TCO objective function for an LRPIF that captures real-world requirements from retail logistics and technical limitations of ECVs. To solve large-size instances and allow for real-world application, we have designed an efficient matheuristic that makes strategic investment decisions with exact algorithmic components, and uses a metaheuristic to solve the remaining operational subproblems. We have presented a real-world case study and developed a technology roadmap to forecast future vehicle, charging station, and battery scenarios for both technological and cost developments. Based on this roadmap, we have performed extensive sensitivity analyses which were further enriched by considering different network characteristics, finally providing results on 168 different scenarios. Our results allow for the derivation of the following managerial key insights.

For certain application cases, ECVs are already on the verge of breaking even: The TEDi case has revealed that the utilization of ECVs in mid-haul logistics networks may be worthwhile, even

under current cost and technical conditions. Specific network characteristics work in favor for the competitiveness of ECVs. Analyzing logistics networks with similar characteristics may reveal comparable results. However, for the majority of other network structures, the results are less favorable under the current conditions.

A disruptive fleet mix change yields the highest cost savings: In all investigated scenarios, ECVs, if any, are deployed in the first two planning periods, independently of the underlying network structure or of the available vehicle type or the charging technology in order to increase operational cost savings. ECV investments in the second time period only occur if this leads to reduced investment costs in charging infrastructure or to better route utilization in the case of changing customer patterns.

The next generation of ECVs allows for a viable (partial) deployment of ECVs in mid-haul logistics networks: As the results for the Tesla Semi scenarios show, a partial electrification of logistics fleets always yields a cost-optimal solution. However, in none but one case, a full electrification is cost-optimal since there are always some less utilized routes which favor the use of ICEVs.

Electrifying mid-haul logistics networks allows to leverage significant ecological improvements: The electrification of mid-haul logistics networks may reduce CO₂ emissions by a factor between 55.4% and 95.7%, depending on the electricity mix. Under present conditions, these solutions yield significant additional costs. However, the next generation of ECVs allows for a cost-optimal partial electrification that still leverages emission savings between 53.9% and 86.6%.

The cost for charging infrastructure heavily affects the economic viability of ECVs under current technological conditions: As can be seen in the eMoss scenarios, under current conditions where en route recharging is necessary due to limited driving ranges, only single network structures with low charging station costs allow for an economically viable electrification of mid-haul logistics fleets.

Certain network characteristics favor the deployment of ECVs: Since the viability of ECVs depends highly on their utilization, mid-haul logistics networks that are operated in a two-shift mode appear to be particularly suitable for the deployment of ECVs. Conditions that often favor the deployment of ECVs lie in varying and more fragmented customer demands once the driving range is sufficiently large.

A final comment in relation with these insights is in order. While our insights and findings represent a natural first step of studies for mid-haul logistics networks, all insights clearly remain case sensitive and must be treated as such. While these findings may reveal tendencies for retail mid-haul logistics networks, significantly different tendencies may apply to cases in which the customer demand remains stochastic and fluctuates on a shorter time scale. Developing a framework capable of providing similar analyzes for networks with stochastic demands remains a central step for a major contribution for future research. Additionally, investigating the use of battery swapping stations could reveal additional interesting insights. The framework presented in this paper can readily be applied to this setting. Moreover, considering public charging stations for recharging purposes may reduce the impact of charging station cost on the viability of ECVs. However, so far logistics fleet operators abstain from integrating public charging infrastructure into their planning to avoid operational disruptions, which may happen in case of blocked stations or vandalism.

Acknowledgements

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Appendix A. Subproblem metaheuristic

To account for the interdependencies of our planning components, an efficient routing component that provides high quality solutions within few iterations is crucial to generating a good solution within reasonable runtimes. Fig. A.14 shows the pseudocode for a single step of our metaheuristic which is embedded within the general algorithm, as explained in Section 3. We use a hybrid of large neighborhood search (LNS), local search (LS), and dynamic programming (DP). The DP component is adapted from Schiffer and Walther (2018a) to efficiently schedule charging stops. We use this DP component only if the cost of the temporal solution σ' does not exceed $(1 + \delta)$ times the cost of the best known solution σ^* .

In the following, we detail the general solution representation and handling within our algorithm as well as the components of the LNS and the LS.

A.1. Solution representation and handling

If a subproblem covers multiple time steps, we use clusters, each representing a time step, within our solution. Accordingly, customers can only be moved within their cluster. Investment decisions for vehicles and charging stations may affect multiple time steps, i.e., all clusters succeeding to the investment. Hence, our clusters exhibit a hierarchical order that allows to integrate such investment decisions.

To escape from local optima, we allow for the temporary acceptance of infeasible solutions. We consider a temporary solution σ' , a current solution σ , and a so far best found solution σ^* . All of these solutions may remain infeasible during our search. Besides, we consider a temporary feasible solution σ'_f and the best known feasible solution σ_f^* . At each search step, we handle these solutions as follows (see Fig. A.14, lines 5–12). We forward the temporary solution σ' to the current solution σ and the so far best known solution σ^* after each search step, if this yields an improvement. We then create a temporary feasible solution σ'_f out of σ' using the state-of-the-art method of Vidal, Crainic, Gendreau, and Prins (2014). This feasible solution is included in a solution pool ϑ that we use for a set partitioning component when taking the investment decisions (see Section 3.2). Analogously, we forward σ'_f to σ_f^* , if it yields an improvement.

To handle infeasible solutions during our search, we use a generalized cost function (A.1) (cf. Schiffer & Walther, 2018a). Besides considering the solution costs $\lambda(\sigma')$, it prices constraint violations by adding penalty terms for freight (FR), time window (TW) and battery capacity (FL) violations. As in Gendreau, Hertz, and Laporte (1994) penalty terms are each multiplied by a self-adaptive weight (α, β, γ) . During the search, the weights

```

1:  $\sigma' \leftarrow \text{destroyAndRepair}(\sigma')$ 
2:  $\sigma' \leftarrow \text{localSearch}(\sigma')$ 
3: if  $\lambda(\sigma') \leq (1 + \delta)\lambda(\sigma^*)$  then
4:    $\text{dynamicProgramming}()$ 
5: if  $(\lambda(\sigma') < \lambda(\sigma))$  then
6:    $\sigma \leftarrow \sigma'$ 
7:   if  $(\lambda(\sigma') < \lambda(\sigma^*))$  then
8:      $\sigma^* \leftarrow \sigma'$ 
9:    $\sigma'_f \leftarrow \text{generateFeasibleSolution}(\sigma')$ 
10:   $\vartheta \leftarrow \sigma'_f$ 
11:  if  $\text{feasible}(\sigma'_f)$  and  $(\lambda(\sigma'_f) < \lambda(\sigma_f^*))$  then
12:     $\sigma_f^* \leftarrow \sigma'_f$ ,  $i^i \leftarrow i$ ,  $\text{updateLB}()$ 

```

Fig. A1. Pseudocode overview of the `solveSubproblem()` routine.

vary between $(\alpha_{\min}, \beta_{\min}, \gamma_{\min})$ and $(\alpha_{\max}, \beta_{\max}, \gamma_{\max})$. After initializing $(\alpha = \alpha_0, \beta = \beta_0, \gamma = \gamma_0)$, we divide or multiply each penalty factor by ω , depending on whether or not a violation has occurred in the last ι^{pen} iterations of the search for the respective constraint. The penalized objective is

$$\lambda(\sigma) = \lambda(\sigma') + \alpha FR + \beta TW + \gamma FL. \quad (A.1)$$

We create an initial solution with a modified savings algorithm. First, we construct back-and-forth routes for all customers. Then, we sort potential merge moves in non-increasing order of their respective cost savings. We merge tours according to this savings list until no positive cost saving remains. While merging, we forbid violations of the vehicle's freight capacity, but we allow for time window and battery capacity violations. Finally, we use our local search to improve the initial solution once.

A.2. Destroy and repair phase

At each iteration, we destroy the current solution σ with a destroy operator and we then apply a repair operator to create a new solution. This destroy and repair phase differs from current LRPIF approaches, as it only affects the route patterns by removing and inserting customers and not the charging station investments. Both destroy and repair operators are randomly chosen at each iteration. To remove customers, we use a worst remove (Ropke & Pisinger, 2006), a related remove (Pisinger & Ropke, 2007), and a modified Shaw remove as well as a station vicinity remove (Goeke & Schneider, 2015) operator. To insert customers, we use a sequential insertion operator (Hiermann et al., 2016) and we design a perturbed version of this operator that randomly swaps the order of 20% of the candidate insertions.

A.3. Local search

In our local search, we evaluate a composite neighborhood structure. Additionally, we have to account for the different vehicle types and characteristics. Hence, we design two versions of each operator. In the homogeneous version, the operators are limited to nodes and routes of a single vehicle type, whereas in the heterogeneous version nodes between ICEVs and ECVs can be exchanged.

Our neighborhood structure comprises a homogeneous 2-opt* (Potvin & Rousseau, 1995) and an heterogeneous Or-opt (Or, 1976) operator. Additionally, we use a heterogeneous exchange and a heterogeneous relocate operator (Savelsbergh, 1992) in inter- and intra-route fashion. For ECV routes, we additionally design a stationInsertionRemoval operator that inserts or removes visits to charging stations. We evaluate these operators with a best-improvement criterion in a decent neighborhood fashion. We consider only the 20% closest neighborhoods for each vertex and enlarge our evaluation to the complete neighborhood when no further improvement can be reached. This technique significantly helps to speed up our algorithm without worsening its solution quality.

Appendix B. Parameter setting and computational validation

We carried out all numerical experiments on a desktop computer with an Intel Core i7 3.60 gigahertz and 16 gigabyte RAM running Ubuntu 16.04 LTS. We implemented our algorithm as a single thread code in C++. To identify a suitable parameter setting, we applied the method of Ropke and Pisinger (2006) on a random subset of our instances.

Table B.6 shows the final parameter setting for the relevant parameters. This table contains the number of iterations after which the investment decision is optimized (ι^{opt}) and the number of iterations for which a subproblem is solved (ι^{max}) from our global

Table B1
Parameter setting for our algorithm.

ι^{opt}	ι^{max}	ι^{pen}	δ	ω	$\alpha_{\max}, \beta_{\max}, \gamma_{\max}$	$\alpha_{\min}, \beta_{\min}, \gamma_{\min}$	$\alpha_0, \beta_0, \gamma_0$
100	2000	50	0.1	1.05	10	10000	100

algorithm, as well as the number of iterations after which penalties are updated (ι^{pen}), the factor for the dynamic programming corridor (δ), the penalty update factor (ω), the lower and upper bound for penalty weights ($(\alpha_{\min}, \beta_{\min}, \gamma_{\min})$, $(\alpha_{\max}, \beta_{\max}, \gamma_{\max})$), and the initial values for penalty weights ($\alpha_0, \beta_0, \gamma_0$).

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ejor.2020.09.054.

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